

a. Already existing OCR engines:

Tesseract 5: an open-source OCR engine that uses a neural network to extract text from images. It supports 100+ languages and works well mostly with clean, clearly printed documents. **Pre-processing:** converts to grayscale, upscales small images to ~2000px, doubles contrast, sharpens, then hard-binarizes to pure black-and-white.

RapidOCR: a fast, lightweight OCR engine built on top of PaddleOCR (Baidu's OCR framework) but repackaged to run faster and without needing the full PaddlePaddle deep learning framework, optimized for production environments where speed and throughput matter without sacrificing accuracy. **Pre-processing:** Keeps the image in color (RGB), applies a mild contrast boost (1.5x), converts to a NumPy array.

EasyOCR: a Python-based OCR library that uses deep learning to extract text from images. It's built on PyTorch and it's meant to handle real-world images better than Tesseract. **Pre-processing:** Same as RapidOCR but adds a sharpness boost (2.0x) on top of the contrast boost.

Printed Rx example:

Patient Name: John Smith
 Address: 400 E 3rd Street
 Duluth, MN 55804
 Rx Amoxicillin 250 mg tablets # 42
 TT tablets p.o. T.i.d. X 7 days
 Do Not Refill ☒ (Sign) M.D.
 Refill Times D.E.A. Number —
 Date 10/3/00 Print Last Name Johns

Tesseract 5	RapidOCR	EasyOCR
Patient Name: John smith Address: 400 E 3rd Street Duluth, MN 55804 Rx Amoxicillin 250 mg tablets TT tablets p.o. T.i.d. x 7 days Do Not Refill __X_ (sign) wc. Refill Times D.E.A. Number — Date 10/3/00. Print Last Name Johns _	PatientName:John Smith Address: 400E 3rd Street Duluth.MN 55804 Rx Amoxicillin 250 mg tablets # 42 TT tablets p.o. T.i.d, X 7 dayS cti Do Not Refill (Sign) M.D. Refill Times D.E.A. Number Date 10/3/00 Print Last Name _Johns	Patient Name: John Smith Address; 4QQE 3rd Street Duluth MN 55804 Rx Amoxicillin 250 mg tablets #42 TT tablets p.o Ti.d X days k Do Not Refill M.D. Refill Times DEA_ Number Date 10300 Print ast Nane Johns actic OT
		Didn't extract duration of treatment

Handwritten Rx example 1:

**Medical prescription by
Klipa Medical Center**
Lubeckweg 2, 9723HE
Groningen, The Netherlands

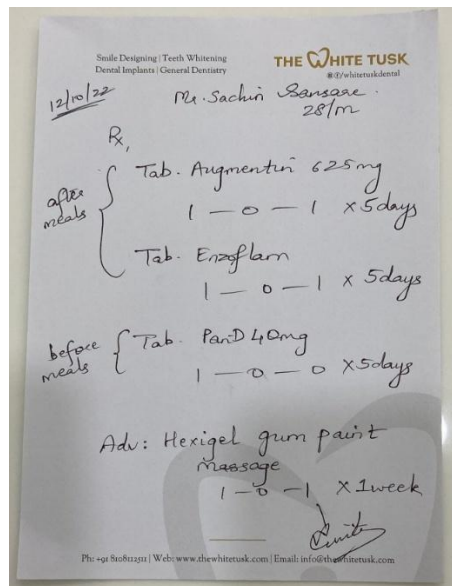
Patiënt name: Mario Carlos van der Vaart
Patiënt age: 38
Date: 20-01-2025

Rx Paracetamol - 50mg
Ibuprofen - 150mg

Signed by:
Dr. Bill Andrew Balmer

Tesseract 5	RapidOCR	EasyOCR
Medical prescription by Klippa Medical Center Lubeckweg 2, 9723HE Groningen, The Netherlands Patiënt name: Patient age: Date: K Signed by: Mario CavloS van der Vaart 49 10-01-1015 Paracetamol - 50m, Ibuprofen - 150m Or. Bill Andrew Balmer	Medical prescription by Klippa Medical Center Lubeckweg 2, 9723HE Groningen, The Netherlands Patient name: Mario Carlos van der Vaart Patient age: 38 Date: 20-01-2025 Rx Paracetamol - 50mg Ibuprofen - (50mg Signed by: Dr. Bill Andrew Balmer	Medical prescription by Klippa Medical Center Lubeckweg 2, 9723HE Groningen; The Netherlands Patient name: Mario Carlos van der Vaart Patient age: 38 Date: 10-01-1015 Paracetamol Ibuprofen Signed by: Dr. Bill Andrew Balmer 50mg 150wg
- Wrong units - One wrong drug name	- One wrong drug name - (50 instead of 150	- Dosages extracted at the end (can't identify which belongs to which drug) - Wrong unit (wg) - 150 not 150 & 50 not 50

Handwritten Rx example 2:



Tesseract 5	RapidOCR	EasyOCR
Smile Designing Teeth Whitening THE WHITE TUSK. Dental Implants General Dentistry @O/whitetuskdental [22 Me Sachin Beansrae me Tab - fragmento. 625d (—o— / «La “Tab- Enadplacn —o— << yee “Fak - faxD 4	THEWHITETUSK SmileDesigning Teeth Whitening DentalImplantsGeneralDentistry @/whitetuskdental MSachun Sansaae 12/10 /22 281m Tab. Augmentn 625mg meals 1 x 5days Adv: Hexigel gum Pant massage 1--1x1week Ph: +91 8108112511/Web:www.thewhitetusk.com Email:info@thewhitetusk.com	Smile Designing Teeth Whitening THE CJHITE TUSK Dental Implants General Dentistry @O/whitetuskdental NL Sachn LSansoae 28/m R Tab - Aujnentu 6 25 m Enzgbrn fTal FanDD 4 0 Adu: Hexiqel qurn 8 (eagage Lweek Ph: +91 810811251 Web: www.thewhitetusk.com Email: info@thexhitetusk com12o/22 ~fl zdv8 Tal . Sda8 Oj Nefct xsda8 Pan -
No correct extractions	Only got one drug name and one dosage correct	All drug names wrong, no dosage or frequency extracted

Conclusion:

- The engines performed fairly well on the printed Rx, but badly on the handwritten ones. If any of them were to be used for printed Rx, a parser would be needed to extract the relevant information
- Can not read symbols indicating schedule
- Did not extract the notes “before/after meals”
- In most cases, the user will end up having to manually correct/enter the correct details

If we were to fine-tune any of the models, a feasible dataset containing many prescriptions & labeled ground truth would be needed, however no such datasets were found.

The available datasets:

Name	Description	Limitations
Doctor’s Handwritten Prescription BD dataset https://www.kaggle.com/datasets/mamun1113/doctors-handwritten-prescription-bd-dataset	Contains images of 78 drug names, cropped from medical prescriptions	Only 78 drug names (mostly trade names) ∴ model cannot recognize drugs outside the training set. Each image only has the specific medicine name not the full prescription ∴ model never learns to locate and extract a drug name from surrounding text. Even if drug name is correctly extracted, it would still fail to extract dosage and timing instructions.
medical-prescription-dataset https://huggingface.co/datasets/chinmays18/medical-prescription-dataset	1,000 synthetic PNG images of prescriptions	All prescriptions are in the same handwriting ∴ model overfits to the one font's visual style and fails to generalize to real-world prescriptions.
Doctors prescriptions handwriting Computer Vision Dataset https://universe.roboflow.com/daffodil-international-university-s5vpr/doctors-prescriptions-handwriting	500 one-line images from multiple handwritten prescriptions, with bounding boxes on each character	Only 500 images so it’s very small dataset for training an OCR model. No sequence context since characters are detected individually, so we would need to reconstruct each word in the prescription.
Handwritten Medical Prescriptions Collection https://www.kaggle.com/datasets/mehaksingal/ill-egible-medical-prescription-images-dataset	Diverse range of medical prescription images obtained from various sources	Images have no labeled ground truth, so the model cannot learn what text corresponds to what region, making supervised training impossible without manually annotating the entire dataset first, which is not doable due to the nature of doctors’ Rxs being very difficult to read and the lack of context.

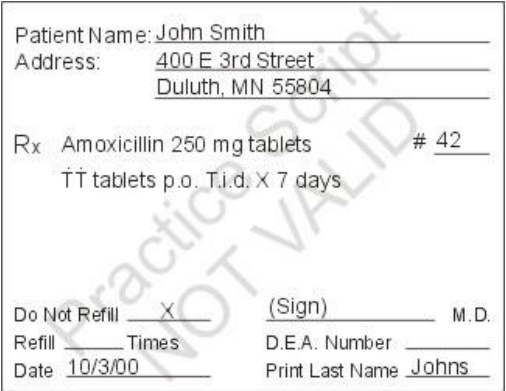
b. Google's ML kit:

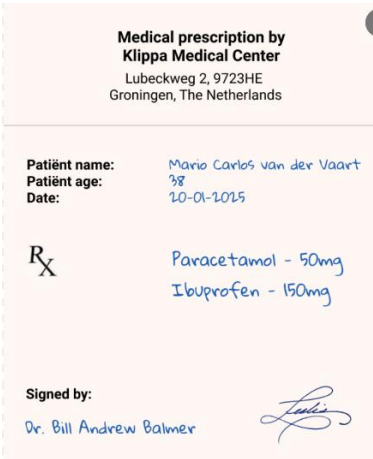
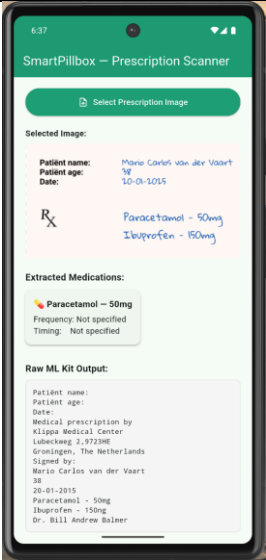
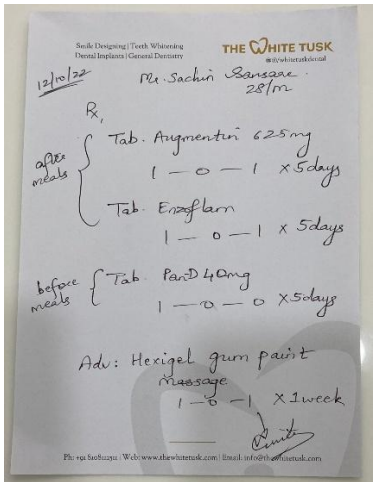
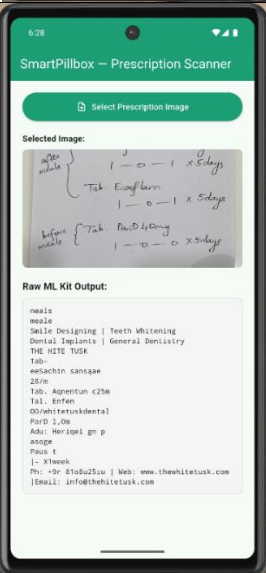
A machine learning software development kit that makes it easy to integrate ML models into mobile apps. We would be using the TextRecognizer engine to extract the information from the Rx using OCR.

Application flow:

1. An image is chosen by the user
2. InputImage.fromFile() takes that file path, reads the image from storage into memory, and packages it into an InputImage object (a ML Kit internal format for representing images)
3. The InputImage object is passed to the TextRecognizer
4. It returns raw text in one long string with line breaks in it
5. Parser loops through every line and locates a capitalized word followed by a dose unit
6. Extracts four fields: drug name (first capitalized word) and dose (number plus unit), frequency (checked against a fixed list of known phrases like "once daily) and timing (checked against a different list: "morning", "evening", etc.)
7. Result is built from extracted fields and displayed

Examples:

		Failed to extract frequency due to not understanding the medical abbreviation (T.i.d = ter in die = three times daily)
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		Extracted only one drug name; failed on the second due to misreading mg as ng
		Extracted none of the handwritten words correctly

Strengths	Limitations
No internet connection required	It has no semantic understanding as it just reads characters, so it does not have any context on anything in the prescription
Completely free	Parser breaks as soon as an unexpected format is used and it would be impossible to hard code every single possible format / phrasing / abbreviation
Fairly fast since image never leaves the device	ML Kit decides where lines begin and end based on spatial layout, so if a medication entry wraps across two lines, the parser has to guess how to reassemble it, leading to inaccuracies
No dependency on an external API means no downtime risk from third-party outages	

Conclusion: the output is unpredictable and the best-case scenario would be a kind of close output that the user still has to manually double-check and correct. Also, based on these examples, using it for a handwritten Rx would give terrible results.

c. Qwen2.5-VL:

The Qwen 2.5 VL 72B Instruct model is a vision language model with 72 billion parameters fine-tuned to follow instructions and have conversations, which makes it perfect for breaking down a doctor's prescription, since it can actually read the document visually and use its medical knowledge to explain what's in it.

Application flow:

1. User picks an image & clicks on extract medications button
2. App reads the image file and converts it to a base64 string to represent binary image data as plain text so it can be sent over HTTPS
3. The base64 string is sent to OpenRouter's API, with the prompt and is then routed to the Qwen 2.5 VL 72B Instruct model
4. Model reads the prescription and extracts the fields mentioned in the prompt and returns a JSON string
5. JSON string is cleaned up from any markdown the model might add and converts it into a list of Medication Dart objects using the model defined in medication.dart
6. Each Medication object becomes a card on the review screen, with every field pre-filled with what the model extracted, but they're all editable TextFormFields, so the user can correct any mistakes

Prompt:

```
Extract all medications from this prescription image.

For each medication, return:
- Drug name
- Dose (e.g. "500mg", "1g")
- Frequency (times per day, e.g. "3 times daily")
- Treatment duration (e.g. "5 days", "1 week") – null if not written
- Special instructions or conditions (e.g. "take with food", "only if fever") – empty string if none
- If a dosing schedule is explicitly written in morning-noon-evening-night format (e.g. 1-0-1-0), extract it as:
  { "morning": 1, "noon": 0, "evening": 1, "night": 0 }
  Otherwise set "schedule" to null – do NOT infer from frequency.

Return ONLY valid JSON. No markdown, no explanation.

Schema:
{
  "medications": [
    {
      "name": "",
      "dose": "",
      "frequency": "",
      "duration": "",
      "schedule": { "morning": 0, "noon": 0, "evening": 0, "night": 0 },
      "notes": ""
    }
  ]
}
```

Printed Rx example:

The image shows a printed prescription form on the left and a mobile app interface on the right. The printed form contains the following information:

Patient Name: John Smith
Address: 400 E 3rd Street
Duluth, MN 55804

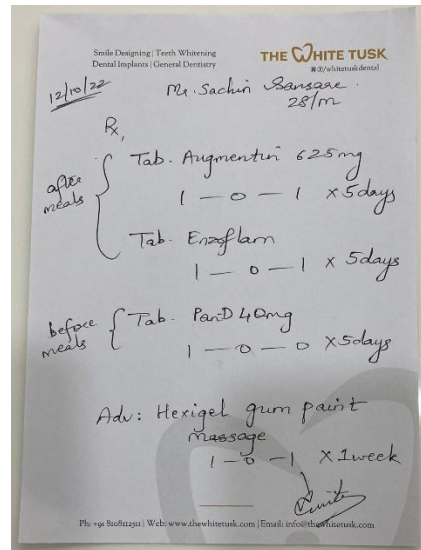
Rx Amoxicillin 250 mg tablets # 42
TT tablets p.o. T.i.d. X 7 days

Do Not Refill X (Sign) _____ M.D.
Refill _____ Times D.E.A. Number _____
Date 10/3/00 Print Last Name Johns

The mobile app interface, titled "Review Medications", shows the following fields:

- 1 Medication 1
- DRUG NAME: Amoxicillin
- DOSE: 250 mg
- FREQUENCY: 3 times daily
- DURATION: 7 days
- NOTES / INSTRUCTIONS: Not detected
- DOSING SCHEDULE: No schedule extracted - tap to add
- Confirm & Save button

Handwritten Rx example:



1 Medication 1

DRUG NAME
Tab. Augmentin

DOSE
625mg

FREQUENCY
2 times daily

DURATION
5 days

NOTES / INSTRUCTIONS
after meals

DOSING SCHEDULE
Morn. 1 Noon 0 Eve. 1 Night 0

2 Medication 2

DRUG NAME
Tab. Enzoftam

DOSE
Not detected

FREQUENCY
2 times daily

DURATION
5 days

NOTES / INSTRUCTIONS
after meals

DOSING SCHEDULE
Morn. 1 Noon 0 Eve. 1 Night 0

3 Medication 3

DRUG NAME
Tab. Pan-D

DOSE
40mg

FREQUENCY
once daily

DURATION
5 days

NOTES / INSTRUCTIONS
before meals

DOSING SCHEDULE
Morn. 1 Noon 0 Eve. 0 Night 0

4 Medication 4

DRUG NAME
Hexigel gum paint massage

DOSE
Not detected

FREQUENCY
2 times daily

DURATION
1 week

NOTES / INSTRUCTIONS
Not detected

DOSING SCHEDULE
Morn. 1 Noon 0 Eve. 1 Night 0

Strengths	Limitations	Potential improvements to limitations
Structured and accurate output from a single call	Requires internet connection	User can be prompted to retry when connection is restored
Understands the image contextually, not just raw text extraction, so it can handle non-standard layouts	Response time varies depending on the image, but overall, it is slower than on-device processing	Image compression before sending
Can edit the prompt as necessary to include/remove fields	API key is exposed in the .env file so anyone who decompiles the Android Package Kit can extract it	Move the OpenRouter call into a Cloud Function
Users will not need to correct as many fields as with other approaches	Cost increases with scaling, but is relatively cheap compared to other models like Claude 3.5 Sonnet or GPT-4o	Hashing and caching the images to return the same cached result in the case of the same image is sent twice. Otherwise, there isn't much of a solution
No on-device ML overhead since all the heavy processing happens in the cloud		

Conclusion: overall showed the best accuracy at extracting prescriptions information so has the least need for user involvement and most of the limitations have fixes, unlike the Google ML Kit.